Kaung Sithu

ALIGNMENT-GUIDED RESIDUAL MODELING FOR LONGITUDINAL MEDICAL VISUAL QUESTION ANSWERING

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ALIGN-VQA: ALIGNMENT-GUIDED RESIDUAL MODELING FOR LONGITUDINAL MEDICAL VISUAL QUESTION ANSWERING

by

Kaung Sithu

A Thesis Submitted in Partial Fulfillment of the Requirements for the Degree of Master of Engineering in Data Science and Artificial Intelligence

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I, Kaung Sithu, declare that the research work carried out for this thesis was in accordance with the regulations of the Asian Institute of Technology. The work presented in it are my own and has been generated by me as the result of my own original research, and if external sources were used, such sources have been cited. It is original and has not been submitted to any other institution to obtain another degree or qualification. This is a true copy of the thesis including final revisions.

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ABSTRACT

Longitudinal image comparison is fundamental to radiological decision-making, requiring clinicians to assess the temporal evolution of findings across visits. Medical Difference Visual Question Answering (Diff-VQA) formalizes this process, however, the current methods for Diff-VQA often rely on direct feature subtraction or simple fusion. These strategies assume inherent spatial correspondence, a premise frequently violated by stochastic acquisition variability and non-rigid anatomical deformations in clinical practice. The author proposes ALIGN-VQA, a novel alignment-guided residual modeling architecture for robust temporal reasoning. Thisg approach introduces three key innovations: (i) Cross-Image Difference Attention (CIDA): Aligns prior-visit features to the current scan via semantic reconstruction, ensuring misalignment-tolerant comparison, (ii) Learnable Residual Gating: Captures precise deviations between aligned historical and current representations to extract evolution-aware features, (iii) Question-Guided Difference Tokenizer (QDT): Selectively pools question-relevant temporal changes for targeted reasoning. Evaluation on the Medical-Diff-VQA benchmark, ALIGN-VQA shows comparable or better performance compared to state-of-the-art methods, with notable improvements in complex reasoning categories. Further, the proposed model ALIGN-VQA is a lightweight architecture making it amenable for edge deployments.

4

CONTENTS

15

7

LIST OF TABLES

Tables **Page**

LIST OF FIGURES

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The automated interpretation of longitudinal radiological images, a cornerstone of clinical practice, has recently been formalized under the task of Difference Visual Question Answering (DiffVQA). The release of Hu et al. (2025) was a pivotal moment, enabling the development of models to answer nuanced questions about temporal changes in patient scans. The subsequent evolution of state-of-the-art models has revealed a consistent, yet resource-intensive, paradigm.

Initial architectural innovations, such as the graph-based representations in Hu et al. (2023) and the dedicated residual encoder in Lu et al. (2024), established methods for processing paired image features. However, these models operate on dense visual feature maps, a computationally expensive approach. Recognizing the need for stronger representations, Cho, Kim, Shin, Cho, and Shin (2024) demonstrated that extensive pretraining on large-scale natural and medical corpora significantly boosts performance, but at the cost of being highly data-hungry and dependent on external reports. More recently, retrieval-augmented models like Yung et al. (2024) and powerful backbones like the Swin Transformer in the Marhuenda, Obrador-Reina, Aas-Alas, Albiol, and Paredes (2025) have pushed performance further. While these models are highly effective, they represent a "brute-force" methodology: they rely on massive model scale, computationally expensive database searches, or the processing of thousands of visual tokens from the entire image to implicitly learn the difference signal.

1.2 Statement of the Problem

The dominant paradigm in Difference VQA is fundamentally inefficient and lacks explicit clinical reasoning. Current models are computationally expensive, requiring the processing of dense visual information from two full images, even when the question pertains to a small, localized change. This brute-force approach is not only wasteful but also fails to explicitly model the directionality of clinical findings, making it difficult to distinguish between concepts like missing or new abnormalities. Furthermore, this reliance on large, complex models without learning gating mechanism makes them prone to capturing non-pathological changes that are not relevant, a critical failure for any clinical support tool. The field lacks a model that is computationally efficient, explicitly models the direction of change, and filters out non-pathological visual noise and static anatomy

1.3 Research Questions

Following the problem, There are three research questions:

1. Can longitudinal, difference-focused questions be accurately answered using only a handful of question-guided difference tokens, instead of processing all image patches?
2. Does the cross image difference attention help the model distinguish the direction of change such as missing or new abnormalities?
3. Does the application of a learnable residual gating mechanism impact the model's ability to capture precise pathologies?

1.4 Hypotheses

55 There are three hypotheses to answer the research question:

34
1. A small set of sparse, question-guided difference tokens is sufficient to achieve state-of-the-art performance, leading to a dramatic increase in computational efficiency.
2. Aligning reference image to current image yields more accurate representation of disease evolution than direct substraction.
3. Applying a learnable gating mechanism to residual features effectively filters out non-pathological noise, resulting in higher diagnostic accuracy.

17
1.5 Objectives of the Study

The main objective is to design and develop lightweight, difference aware medical VQA system that can provide faithful visual evidence to support its answer while being compute efficient and not requiring retrieval databases, radiology reports or expert annotations. The sub-objectives are:

1. To validate the effectiveness of a Question-Guided Diffference Tokenizer (QDT) at distilling visual information into a sparse set of tokens.
2. To evaluate the impact of an integrated Cross Image Difference Attention (CIDA)
43 on the model's ability to reason about the direction of clinical change.
3. To quantify the impact of the learnable residual gating mechanism in isolating precise pathological deviations and extracting evolution-aware features.

1.6 Methodology

This work introduces ALIGN-VQA, a novel, lightweight architecture that shifts from a brute-force comparison to an alignment-guided residual modeling paradigm. The methodology is centered on three key innovations designed to handle the stochastic acquisition variability and non-rigid anatomical deformations inherent in longitudinal imaging. First, to overcome the limitations of spatial misalignment between scans, the CIDA module semantically aligns prior-visit features to the current scan via cross-attention before computing differences. This aligned representation is then passed through a Learnable Residual Gating mechanism, which adaptively suppresses acquisition noise to extract precise, evolution-aware residual features. Second, the QDT module utilizes domain-aware text embeddings from ClinicalBERT to evaluate the relevance of these visual deviations by applying cross-attention to selectively pool only the Top-K most clinically salient temporal changes. The vision components learn contextual coherence via a self-supervised Masked Residual Modeling (MRM) task, while an auxiliary multi-label classification head explicitly ground the visual features in 14 thoracic pathologies.

1.7 Scope

In scope:

- The thesis will exclusively use the public Medical-Diff-VQA dataset, focusing on all question categories for all experiments.
- The core of the work is the design, implementation, and evaluation of our novel, lightweight VQA architecture and its specific modules (CIDA, QDT, CFA-Reg, MRM).
- Evaluation will consist of targeted ablation studies and a direct benchmark against a comprehensive suite of SOTA models (EKAID, ReAl, PLURAL, RegioMix, VED).

Out of scope:

- This work will not involve the collection of new clinical data or the execution of clinical trials with human radiologists.
- This thesis will not develop a new, general-purpose medical VQA system.

1.8 Expected Key Findings

- The proposed, ALIGN-VQA, model is expected to achieve performance comparable or superior to current state-of-the-art models with a fraction of the compu-

2

tational cost, demonstrating the viability of an efficiency-focused paradigm.

- The integration of CIDA module will be shown to provide a significant accuracy boost by explicitly handling spatial misalignments and suppressing stochastic acquisition noise, particularly excelling in complex comparative reasoning categories.
- The learnable gating mechanism will be proven to isolate true pathological deviations from visual noise and static anatomy

1.9 Contribution

- This thesis introduces ALIGN-VQA, a novel, lightweight paradigm for DiffVQA centered on efficiency, explicit reasoning, and robustness, shifting the field away from the dominant brute-force, resource-intensive methodologies.
- Three key architectural innovations are developed, CIDA module for misalignment-tolerant semantic reconstruction, learnable residual gating for isolating true evolution-aware features, and a QDT module that selectively pools clinically salient tokens to prevent signal dilution and maximize computational efficiency.
- A comprehensive multi-stage training strategy that incorporates self-supervised MRM, auxiliary disease classification, and counterfactual learning—is introduced. This yields a state-of-the-art framework that is robust, computationally accessible, and self-contained, removing the reliance on massive external radiology report corpora for pre-training.

33

CHAPTER 2

RELATED WORK

This chapter critically examines the existing body of literature to contextualize ALIGN-VQA architecture and the research questions driving this thesis. Longitudinal medical imaging is a cornerstone of radiological diagnosis, requiring clinicians to meticulously track disease progression or stability over time. In response, Difference-aware Medical VQA (Diff-VQA) has emerged as a specialized domain designed to analyze paired longitudinal scans and answer complex clinical queries regarding their differences. However, despite rapid advancements in generative vision-language models, current Diff-VQA architectures remain constrained by several critical methodological bottlenecks.

2.1 Difference Modeling and the Alignment Problem in Longitudinal VQA

Early approaches to this problem, such as Hu et al. (2023), attempt to structure the comparison by representing anatomical regions as nodes in a multi-relationship graph. By extracting features from identical anatomical bounding boxes across two temporal scans, it aims to evaluate disease progression within specific lung regions. While this explicitly incorporates spatial and semantic medical priors, its reliance on a pre-trained Ren, He, Girshick, and Sun (2015) to detect bounding boxes leaves the model vulnerable; inaccurate feature extraction directly results in incorrect comparative predictions. Furthermore, this strict graph structure struggles to adapt to non-rigid anatomical deformations or novel structural changes not predefined in the knowledge graph.

3

More recent models, such as Lu et al. (2024), compute temporal differences by directly subtracting the reference image from the main image to produce a residual input. This residual is processed through an independent encoder, guided by a consistency loss to enforce focus on disparate regions. However, direct subtraction strategies implicitly assume perfect spatial correspondence between the two scans. In clinical practice, this assumption is frequently violated due to stochastic acquisition variability, patient movement, and postural differences between visits. When raw image subtraction is applied to misaligned scans, it generates overwhelming artifact noise rather than isolating true pathological evolution.

2.2 Feature Selection and Computational Efficieny

Cho et al. (2024) created addresses longitudinal VQA by concatenating the flattened feature vectors of both the past and current images into a single, massive sequence. Utilizing He, Zhang, Ren, and Sun (2016), it generates 576 tokens per image, resulting in a dense, highly redundant input matrix that is processed by a 6-layer Transformer encoder-decoder. While PLURAL benefits from large-scale pretraining on external datasets like Johnson, Pollard, Mark, Berkowitz, and Horng (2024) and T.-Y. Lin et al. (2014), its brute-force processing of all spatial patches limits its efficiency and scalability.

Similarly, Marhuenda et al. (2025) model utilizes a Swin Transformer backbone combined with an Image Differentiation Embedding (IDE) to differentiate the temporal scans. While their model utilizes a lightweight 3-layer decoder, it still fuses and processes the full visual feature maps via cross-attention to generate answers. Other paradigms, such as Yung et al. (2024), attempt to bypass heavy paired-image processing by using Retrieval-Augmented Generation (RAG) to mix-and-match retrieved region features from an external database. While innovative, it requires a continuous, computationally expensive database search for every query and struggles to reason over novel differences not present in the retrieval set.

2.3 Filtering Non-Pathological Noise via Learnable Residual Gating

A persistent weakness in longitudinal medical vision-language models is their inability to distinguish true pathological evolution from stochastic acquisition noise. Generative Diff-VQA models are particularly susceptible to this, as they must predict complex, temporally evolving sequences based on the differences between two scans. When models rely on raw feature subtraction or simple fusion, they treat all visual differences such as changes in patient posture, inspiration effort, or scanner contrast as equally important as actual disease progression. Error analyses of current state-of-the-art models expose the clinical dangers of processing un-gated residual features. For instance, evaluations of Cho et al. (2024) demonstrate instances where the architecture accurately identifies regions of interest but misinterprets the severity sequence, describing an abnormality in reverse temporal order. This occurs because the raw subtracted features are overwhelmed by static anatomical misalignment, confusing the decoder. Similarly, researchers evaluating Marhuenda et al. (2025) noted a critical divergence between automated Natural Language Generation (NLG) metrics and clinical correctness. Models

23

frequently achieved near-perfect scores on metrics like Papineni, Roukos, Ward, and Zhu (2002) and C.-Y. Lin (2004), yet clinical review revealed the model was misinterpreting diagnoses, often incorrectly identifying the resolution of anomalies or falsely flagging new abnormalities. This confirms that without an explicit mechanism to filter out non-pathological visual noise, generative models struggle to isolate and describe true clinical evolution.

2.4 Hypothesis and Positioning

Existing methods indiscriminately process background and irrelevant features, diluting the model's focus on the actual clinical question. This thesis hypothesizes that longitudinal, difference-focused questions can be accurately answered using only a handful of question-guided difference tokens. By introducing a Question-Guided Difference Tokenizer (QDT) that uses cross-attention to score and select only the Top-K most query-relevant visual residual tokens, the proposed model will explicitly filter out irrelevant changes, drastically reducing the computational load for the downstream language model while preventing the dilution of clinically salient deviations.

The limitations of direct subtraction and rigid graph mapping present a critical gap in modeling the direction and accuracy of temporal changes. This research hypothesizes that a Cross-Image Difference Attention (CIDA) module can overcome these limitations by semantically reconstructing and aligning prior-visit features to the current scan's spatial layout via cross-attention before computing residuals. This alignment-guided approach is expected to provide a highly robust mechanism for distinguishing the true direction of change without being derailed by standard acquisition misalignments.

Previous state-of-the-art models in Diff-VQA rely on direct feature subtraction or simple concatenation, operating under the flawed assumption that all pixel-wise differences between two scans are clinically relevant. The architecture of ALIGN-VQA challenges this assumption. By introducing a Learnable Residual Gate, ALIGN-VQA explicitly bridges the gap between raw visual differences and the clinical narrative. Instead of forcing the language decoder to make sense of noisy, un-filtered subtraction maps, our gating mechanism dynamically learns to suppress static anatomy and acquisition artifacts, ensuring that only true, evolution-aware features are passed forward for diagnostic reasoning.

2.5 Comparative Analysis of Methodologies

The following table summarizes the key limitations of the current state-of-the-art models.

Table 2.1

Comparative Analysis of Limitations in State-of-the-Art DiffVQA Models.

Model/Methodology	Computational Inefficiency	Spatial Misalignment Vulnerability Reasoning	High Data Dependency	Susceptibility to Non-Pathological Noise
Hu et al. (2023)	✓	✓		✓
Lu et al. (2024)	✓	✓		✓
Yung et al. (2024)	✓		✓	✓
Cho et al. (2024)	✓	✓	✓	✓
Marhuenda et al. (2025)	✓	✓		✓
Our Proposed Model	*Addressed by QDT*	*Addressed by CIDA*	*Addressed by MRM*	*Addressed by Counterfactual Learning*

Table 2.2

Mapping of Proposed Components to Research Gaps

Proposed Component	Research Gap Addressed	Corresponding Research Question
Question-Guided Difference Tokenizer (QDT)	Computational inefficiency of processing dense visual features.	RQ1: Can longitudinal questions be answered using only a handful of question-guided difference tokens?
Directional Residual Stack (DRS)	Vulnerability to spatial misalignments and implicit assumptions of direct feature substraction	RQ2: Does cross-image difference attention help the model distinguish the direction of change?
Learnable Residual Gating	Susceptibility of generative models to non-pathological visual noise and static anatomy when calculating image differences	RQ3: How does a learnable residual gating mechanism impact the model's ability to capture precise pathological deviations?

CHAPTER 3

METHODOLOGY

3.1 Architectureal Overview

Longitudinal medical image comparison requires models to distinguish true pathological evolution from stochastic acquisition variability, such as patient movement or differences in posture. Conventional Difference-aware Medical VQA methods rely on direct feature subtraction, which implicitly assumes perfect spatial correspondence between scans. This thesis introduces ALIGN-VQA, an alignment-guided residual modeling architecture designed to overcome these spatial misalignment vulnerabilities.

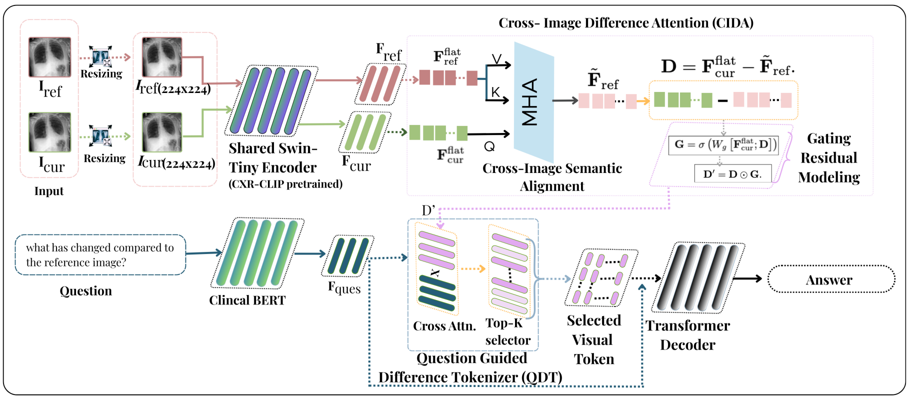

Figure 3.1

High Level Architecture Overview

As illustrated in Figure 3.1, the proposed methodology shifts from a brute-force comparison paradigm to an efficient, explicit reasoning framework. The architecture is composed of three primary functional blocks. Cross-Image Difference Attention (CIDA) and gated residual modeling semantically reconstructs and aligns historical features to the current scan before isolating evolution-aware deviations. Question-Guided Difference Tokenizer (QDT) employs cross-attention to selectively pool a sparse, highly relevant set of temporal changes based on the clinical query. Conditional answer generation and multi-stage training utilizes a lightweight transformer decoder optimized through a highly structured multi-stage training strategy to ensure grounded reasoning.

3.2 Problem Formulation

51

Given a patient's historical reference X-ray I_{ref}, a current follow-up X-ray I_{cur}, and a clinically grounded question Q, the objective of the Diff-VQA system is to generate an accurate natural language answer $\hat{A}$ describing the temporal evolution of the targeted pathology. Instead of mapping inputs directly to a dense difference map, our formulation isolates question-relevant residual deviations to conditionally autoregress the answer.

3.3 Architectural Components

3.3.1 Visual Feature Extraction

31

The visual processing pipeline begins by encoding the paired longitudinal images. Both I_{ref} and I_{cur} are resized to 224×224 and passed through a shared tiny variant of Liu et al. (2021). The shifted-window self-attention mechanism within the encoder enables highly efficient local-to-global context modeling. To inject domain-specific radiographic priors immediately into the feature extraction phase, the backbone is initialized with pretrained weights from You et al. (2023). This process yields two feature maps, which are flattened into token sequences:

$$F_{ref}^{flat}, F_{cur}^{flat} \in \mathbb{R}^{B \times N \times C} \tag{3.1}$$

12

where B is the batch size, C is the channel dimension, and $N = H \times W$ represents the number of spatial patches.

3.3.2 Cross Image Difference Attention

To compensate for spatial misalignments resulting from varying patient positioning and inspiratory effort between visits, we introduce the CIDA module. It utilizes Multi-Head Cross-Attention (MHA) to semantically align the historical features to the coordinate space of the current image. Specifically, the current image features are utilized as queries (Q), while the reference image features serve as keys (K) and values (V):

$$\tilde{F}_{ref} = \text{MHA}(F_{cur}^{flat}, F_{ref}^{flat}, F_{ref}^{flat}) \tag{3.2}$$

This operation reconstructs an aligned historical representation ($\tilde{F}_{ref}$) that is spatially coherent with the current scan, creating a reliable foundation for calculating true temporal differences.

3.3.3 *Learnable Gated Residual Modeling*

Once the features are aligned, the raw residual deviation is computed via element-wise subtraction:

$$D = F_{cur}^{flat} - \tilde{F}_{ref} \tag{3.3}$$

Because medical imaging inherently contains acquisition noise, utilizing the raw residual D directly can mislead downstream modules. Therefore, a learnable gating mechanism is applied. The gate calculates a channel-wise importance filter by concatenating the current features with the raw residual and passing them through a projection matrix W_g and a sigmoid function σ:

$$G = \sigma(W_g[F_{cur}^{flat}; D]) \tag{3.4}$$

The refined, evolution-aware residual features are then extracted via element-wise multiplication:

$$D' = D \odot G \tag{3.5}$$

The resulting tokens $\{D'_i\}_{i=1}^{N}$ encode clean, structured temporal changes.

3.3.4 *Domain-Aware Text Encoding*

The clinical question Q is encoded using Alsentzer et al. (2019), a language model extensively pretrained on clinical notes. This provides a domain-aware question embedding vector $F_{ques} \in \mathbb{R}^C$ that captures the semantic nuance of radiological terminology such as distinguishing between consolidation and atelectasis.

3.3.5 *Question-Guided Difference Tokenization*

Standard Diff-VQA models process all spatial tokens, which dilutes the signal of localized pathologies. The QDT module solves this by selectively pooling only the temporal changes that are relevant to the specific clinical question.Attention weights (α_i) are computed between the question embedding F_{ques} and each visual residual token D'_i using dot-product attention. To enforce extreme token sparsity and computational efficiency, a hard Top-K selection mechanism is applied:

$$T_{vis} = \text{TopK}(\{\alpha_i D'_i\}) \tag{3.6}$$

Based on empirical ablations, K is set to 16, ensuring the model forwards only the most highly discriminative visual evidence to the language decoder.

3.3.6 *Conditional Answer Generation*

The final response generation relies on a lightweight standard transformer decoder. The selected sparse visual tokens T_{vis} are concatenated with the question embedding F_{ques}. The decoder autoregressively predicts the answer $\hat{A}$ step-by-step, conditioned strictly on the question semantics and the isolated, aligned temporal changes.

3.3.7 *Counterfactual Regularizer*

To mitigate the model's tendency to hallucinate, a counterfactual regularizer (CFA-Reg) is introduced. For each training sample (I_{cur}, I_{ref}, Q, A), we generate a logical opposite by either swapping the images (I_{ref}, I_{cur}) or negating a directional term in the question (e.g., "worsened" $\rightarrow$ "improved"). The model performs a forward pass on both the real and counterfactual inputs, generating two distinct answer distributions, P_{real} and P_{cofa}.

3.3.8 *Masked Residual Modeling*

The masked residual modeling (MRM) process is designed to force the model to learn the contextual structure of the difference maps. First, an unlabeled image pair (I_{cur}, I_{ref}) is passed through the vision encoder to generate the concatenated residual map, R. A significant portion of the patches in this map are then randomly masked. The unmasked patches are fed into the QDT's vision backbone, which is tasked with predicting the content of the masked patches. Figure 3.2 shows architecture of masked residual modeling module.

The training of MRM is guided by a reconstruction loss, which measures the dissimilarity between the model's predicted patches and the original, unmasked patches. We use a Mean Squared Error (MSE) loss for this objective.

$$L_{\text{MRM}} = \frac{1}{|M|} \sum_{i \in M} (R_i^* - \hat{R}_i^*)^2$$

Here, M is the set of masked patch indices, R_i is the original patch, and $\hat{R}_i$ is the reconstructed patch.

3.4 Multi-Stage Training Strategy

The model is optimized using a comprehensive four-stage training regimen to ensure model robustness, prevent shortcut learning, and explicitly ground the vision-language

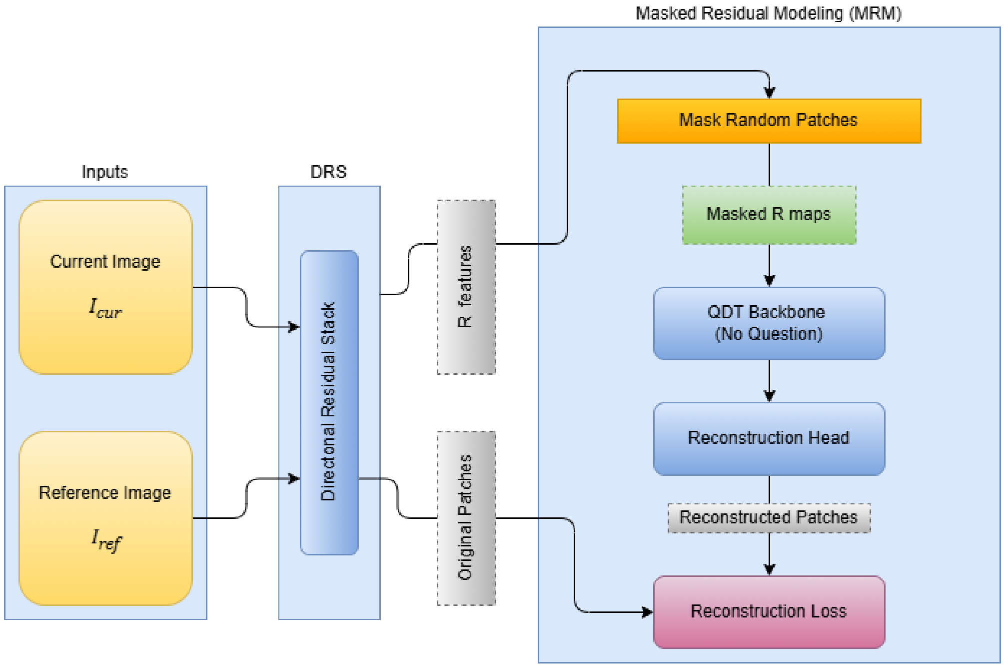

Figure 3.2

Masked Residual Modeling

alignment,

To enforce contextual coherence in the visual representation, 50 percent of the residual tokens are randomly masked in the first stage. A lightweight reconstructor is trained to predict the masked tokens from the visible context using a Mean Squared Error (MSE) objective ($\mathcal{L}_{mrm}$).

In second stage, a multi-label classification head is attached to predict 14 standard thoracic diseases from both the reference and current features using a Binary Cross-Entropy loss ($\mathcal{L}_{cls}$) to explicitly ground the visual features in known pathology

To mitigate spurious correlations and dataset biases and hallucinations, the model applies a combination of hierarchical Kullback-Leibler (KL) divergence and Noise Contrastive Estimation (NCE) in third stage. By contrasting true longitudinal pairs against synthetically perturbed counterfactual pairs, the model disentangles true temporal evolution from visual noise ($\mathcal{L}_{cf}$).

In the last stage, the entire architecture is fine-tuned for the generation task using a standard language cross-entropy loss ($\mathcal{L}_{lang}$) alongside an L1 sparsity regularization penalty on the gating weights ($\mathcal{L}_{gate}$) to force the gate to actively suppress non-essential information. The total loss of the complete training objective is weighted sum of these components.

$$\mathcal{L}_{total} = \lambda_{lang}\mathcal{L}_{lang} + \lambda_{gate}\mathcal{L}_{gate} + \lambda_{mrm}\mathcal{L}_{mrm} + \lambda_{cls}\mathcal{L}_{cls} + \lambda_{cf}\mathcal{L}_{cf} \tag{3.7}$$

3.5 Implementation Details

To ensure full reproducibility, ALIGN-VQA was developed using Paszke et al. (2019). The model features a highly efficient parameter footprint of approximately 182.5M parameters. The QDT module retains a maximum of 64 tokens during early training, which is dynamically pruned to $K = 16$ during fine-tuning. The maximum generated answer length is constrained to 128 tokens. Optimization is performed using the AdamW optimizer with a batch size of 128 and an initial learning rate of 5×10^{-5}, governed by a cosine annealing learning rate scheduler. The training schedule consists of 1 epoch of MRM, 3 warm-up epochs, and 20 epochs of end-to-end VQA fine-tuning. The empirically derived loss weights are set to $\lambda_{lang} = 1.0$, $\lambda_{mrm} = 0.1$, $\lambda_{cf} = 0.1$, and $\lambda_{cls} = 0.5$. Vocabulary mismatch issues during inference are handled dynamically by matching the initialized target vocabulary dimension to the exact embedding size preserved in the saved model checkpoints.

CHAPTER 4
EXPERIMENTS

4.1 Rationale

The primary objective of these experiments is to rigorously evaluate the clinical viability, computational efficiency, and reasoning accuracy of the proposed ALIGN-VQA architecture. Traditional Medical Visual Question Answering (Med-VQA) models often falter when applied to longitudinal tasks because they are vulnerable to spatial misalignments and struggle to isolate true disease progression from harmless acquisition noise. Therefore, the experimental design is motivated by three key goals. First, the author argues that spatial misalignments and stochastic acquisition variability between historical and current scans compromise direct feature comparison. Thus, the Cross-Image Difference Attention (CIDA) module is evaluated on its ability to semantically reconstruct and align the prior-visit features to establish a reliable, misalignment-tolerant baseline. Second, rather than assuming all raw pixel-wise differences are clinically relevant, the author hypothesizes that a Learnable Residual Gating mechanism is required to actively filter out non-pathological visual noise and static anatomy, ensuring that only precise, evolution-aware features are extracted. Finally, because the vast majority of regions in a longitudinal chest X-ray remain unchanged, the author evaluates the Question-Guided Difference Tokenizer (QDT) to determine if selectively pooling a sparse subset of highly relevant tokens enables the model to perform targeted diagnostic reasoning while maintaining a highly efficient, lightweight architecture.

4.2 Dataset

To test these hypotheses, the author utilizes Hu et al. (2025), which is a specialized derivative of the large-scale Johnson, Pollard, et al. (2024). Each sample in this dataset is structured as a longitudinal pair consisting of a historical reference radiograph and a more recent follow-up scan, stored in Digital Imaging and Communications in Medicine (DICOM) format. The x-rays images in Joint Photographic Experts Group (JPEG) format are fetched from Johnson, Lungren, et al. (2024) for faster preprocessing and training. These pairs are accompanied by clinically grounded questions that specifically target temporal changes, such as the progression of pleural effusion or the placement of medical devices. The dataset comprises 164,324 pairs of main and reference chest X-ray images, yielding a total of 700,703 question-answer pairs. This high volume of data allows the model to learn the subtle semantic nuances required to distinguish between common clinical conditions like consolidation, edema, and atelectasis within a comparative context. Following standard benchmarking protocols, the dataset is strictly partitioned into training, validation, and testing sets using an 8:1:1 ratio. The Figure shows the distribution of seven distinct clinical categories

4.2.1 Pathological Distribution

Figure 4.1 illustrates the prevalence of medical findings across the dataset. The distribution exhibits a heavy long-tail characteristic typical of medical datasets. "No Finding" and "Support Devices" are the most prevalent positive labels, reflecting the high frequency of routine exams and Intensive Care Unit (ICU) patient monitoring in Johnson, Pollard, et al. (2024). Among true pathological findings, "Pleural Effusion," "Lung Opacity," and "Atelectasis" occur most frequently. Conversely, critical but less frequent findings like "Fracture" and "Pleural Other" appear significantly less often. This class imbalance highlights the necessity for robust representation learning, particularly the Counterfactual Regularizer in the ALIGN-VQA architecture, to prevent the model from defaulting to high-frequency predictions.

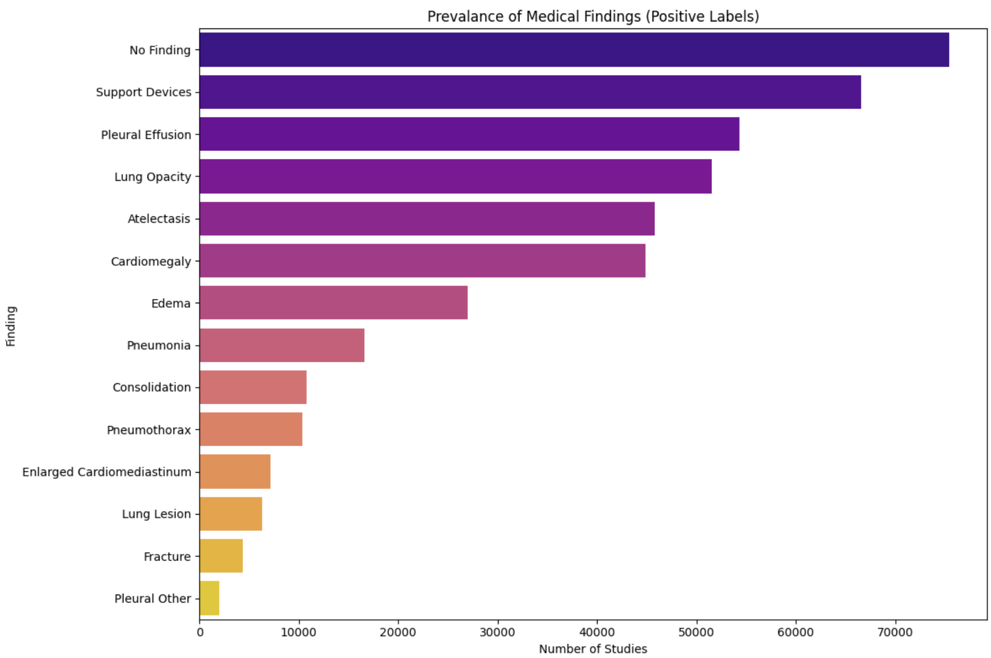

Figure 4.1

Prevalance of Medical Findings

32

4.2.2 Question Type Distribution

The questions are categorized into seven distinct clinical types, as shown in Figure 4.2. The distribution is relatively balanced among the top categories, ensuring a diverse evaluation of the model's capabilities. The "Difference" category is the most frequent accounting for approximately 23 percent of the dataset, followed closely by "Presence", 22 percent and "Abnormality" 21 percent. The remaining categories—Location, Level, View, and Type—make up the rest of the dataset. While the ALIGN-VQA model is trained and evaluated across all seven categories, the primary focus of these experiments remains the "Difference" subset, as it explicitly tests the model's ability to reason about longitudinal spatial and pathological changes between the reference and current scans.

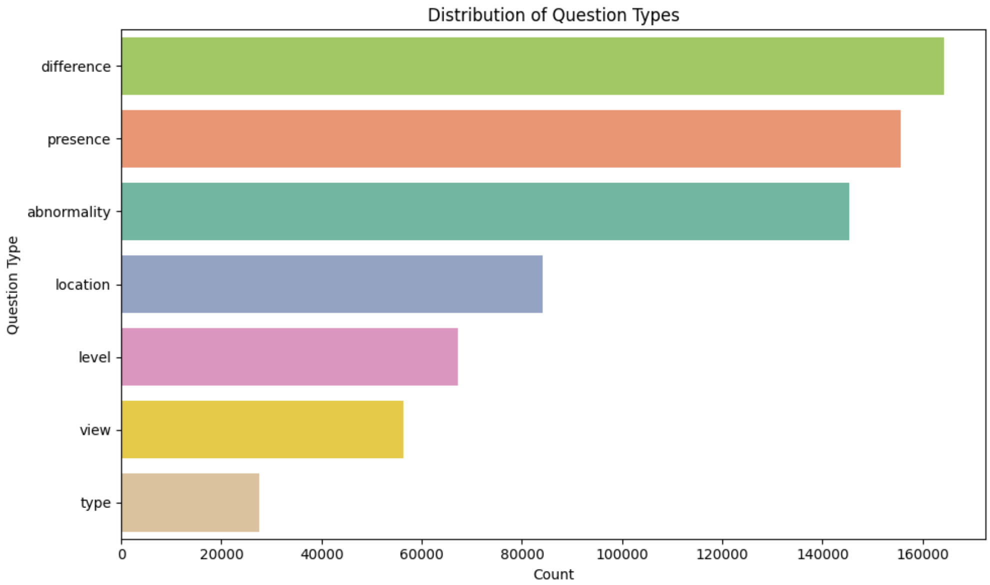

Figure 4.2

Distribution of Question Types

4.2.3 Linguistic Distribution

Figure 4.3 visualizes the distribution of question and answer lengths in characters. The question lengths exhibit a highly structured, multi-modal distribution with distinct peaks around 30, 42, and 50 characters. This rigid structure suggests that the questions were generated using a programmatic or templated approach based on the original radiology reports, rather than free-form human querying. The answer lengths are heavily skewed, with the vast majority of answers being extremely concise, under 25 characters, corresponding to binary "yes/no" answers or short categorical responses. A smaller, secondary distribution of longer answers, 50 to 150 characters, corresponds to the complex, descriptive answers required for the "Difference" and "Abnormality" question types.

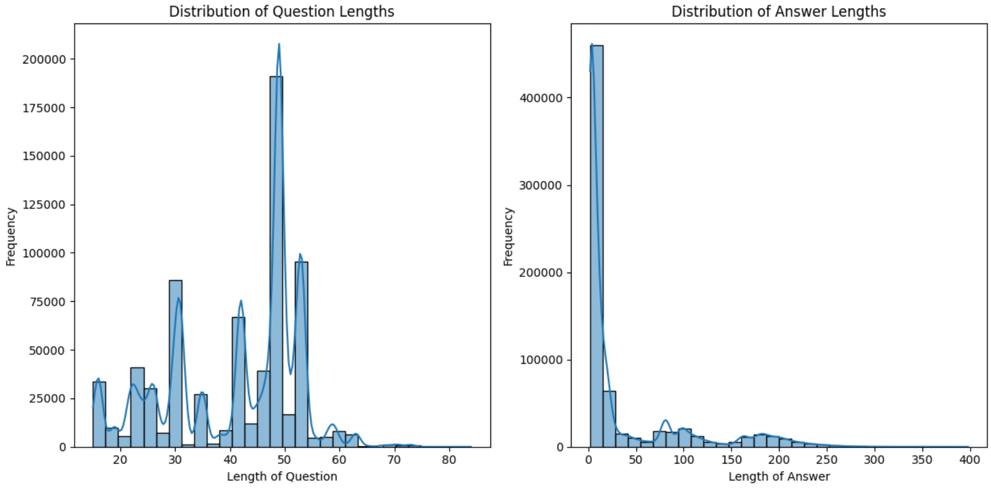

Figure 4.3

Distribution of Question Lengths

4.3 Experiment Setting

4.3.1 Preprocessing

To ensure standardized inputs for the vision-language pipeline, rigorous preprocessing protocols were applied to both the imaging and textual data. The resizing pipeline is engineered to maximize computational efficiency while preserving critical clinical details. Both historical and current chest X-rays were resized to 224×224 pixels to match the input dimensions expected by the tiny varient of Liu et al. (2021). Standard normalization was applied using You et al. (2023) pretraining statistics to ensure stable feature extraction. Clinical questions were tokenized using Alsentzer et al. (2019), with sequences padded and truncated to a maximum length of 48 tokens. To prevent index-out-of-bounds errors and size mismatches during evaluation, a common issue when dataset splits yield varying unique word counts, the inference scripts were programmed to dynamically extract the exact expected vocabulary size directly from the saved model checkpoints prior to initializing the classification head.

4.3.2 Models and Techniques

To benchmark the proposed ALIGN-VQA architecture, its performance is compared against several state-of-the-art models that encompass both general domain change-captioning and specialized medical Diff-VQA. Among the specialized medical models, the author evaluates Hu et al. (2023), a foundational Diff-VQA framework that relies on pre-extracted bounding boxes and an anatomical knowledge graph to model changes.

The author also compares the method with Cho et al. (2024), a computationally intensive transformer framework that utilizes massive external pretraining on natural images and radiology reports to process dense, unpruned feature concatenations. Additionally, the author benchmarks against Yung et al. (2024), a Retrieval-Augmented Generation (RAG) approach that generates pseudo-difference descriptions by mixing and matching retrieved regions from an external database. Furthermore, the author includes two recent generative medical VQA architectures including Lu et al. (2024), which employs a dedicated residual encoder and residual feature alignment to capture image disparities, and Marhuenda et al. (2025), which utilizes Liu et al. (2021) coupled with image differentiation embeddings to distinguish temporal scans prior to decoding.

Against these baselines, the author evaluate the proposed ALIGN-VQA, an alignment-guided residual modeling architecture featuring CIDA, a learnable gating mechanism, and the QDT. Beyond direct comparisons, extensive ablation studies are conducted on ALIGN-VQA by systematically toggling key components. For instance, the CIDA module is disabled to force direct feature subtraction, and the QDT Top-K selection threshold is altered. This allows the author to rigorously isolate and quantify the individual contributions of these novel mechanisms to the overall diagnostic accuracy.

4.3.3 Training Procedure

The model was implemented in Paszke et al. (2019) and trained end-to-end to ensure full reproducibility. All training procedures were executed on NVIDIA RTX6000 to efficiently accommodate the required batch sizes and the computational overhead of the attention matrix operations. The network was optimized using the AdamW optimizer with a batch size of 128. To ensure smooth convergence, the initial learning rate was set to 5×10^{-5} and governed by a cosine annealing scheduler.

The training process strictly adhered to a multi-stage strategy designed to incrementally build the model's reasoning capabilities. This schedule commenced with one initial epoch of self-supervised Masked Residual Modeling (MRM), followed by three warm-up epochs, and concluded with 20 epochs of end-to-end Diff-VQA fine-tuning. The compound loss function was balanced using empirically derived weights, applying a weight of 1.0 to the language generation loss, 0.1 to the MRM loss, 0.1 to the counterfactual loss, and 0.5 to the auxiliary classification loss. Finally, early stopping was implemented based on validation CIDEr scores to continually monitor generalization and prevent overfitting during the fine-tuning phase.

4.3.4 Evaluation Metrics

Because automated metrics often struggle to capture the clinical nuance of medical texts, the evaluation strategy utilizes a robust mix of strict string-matching, semantic similarity, and human validation. For automated assessment, the author reports standard Natural Language Generation (NLG) metrics, including Papineni et al. (2002), Banerjee and Lavie (2005), and C.-Y. Lin (2004), to evaluate sequence overlap and structural fluency. The author places a heavy priority on the Oliveira dos Santos, Colombini, and Avila (2021) metric, as its weighting mechanism explicitly penalizes generic answers and rewards the accurate generation of rare, clinically vital keywords, such as specific pathology names.

CHAPTER 5
RESULTS

It is essential to establish the analytical framework used to evaluate the proposed ALIGN-VQA architecture before detailing empirical findings. The primary objective of these experiments is to rigorously evaluate the clinical viability, computational efficiency, and reasoning accuracy of the model.

To conduct a comprehensive analysis, the author utilized Papineni et al. (2002) to measure precise structural n-gram overlap, C.-Y. Lin (2004) to evaluate the preservation of the chronological clinical narrative, Banerjee and Lavie (2005) to account for valid semantic synonyms, and Oliveira dos Santos et al. (2021) to measure the model's ability to capture high-value, rare diagnostic keywords. Additionally, Zhang*, Kishore*, Wu*, Weinberger, and Artzi (2020) was employed to assess deep semantic similarity. The results are structurally aligned with the three core research questions, analyzing the model sequentially through its forward-pass pipeline: establishing spatial alignment, filtering non-pathological noise, and performing targeted token reasoning.

Table 5.1 presents the quantitative performance of the proposed ALIGN-VQA against current state-of-the-art models on the Hu et al. (2025). The evaluated baselines include early graph-based methods, retrieval-augmented models, dense-feature frameworks, and residual-based generative models. To isolate the contributions of the core architectural innovations, the author conducted extensive ablation studies, the results of which are detailed in Table 5.2. The author evaluated the model by systematically toggling the Cross Image Difference Attention (CIDA) module, Learnable Residual Gating Mechanism (GATE), and varying the Top-k threshold in the Question-Guided Difference Tokenizer (QDT).

5.1 Results Corresponding to CIDA

The first research question investigates how semantic reconstruction via the CIDA module affects the model's robustness to spatial misalignments between historical and current scans. The author hypothesized that establishing a misalignment-tolerant baseline is required before calculating image differences. To evaluate this, the author conducted an ablation study comparing a baseline model utilizing direct feature subtraction against

Table 5.1

Quantitative performance comparison of the proposed model against state-of-the-art methods on Hu et al. (2025).

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr
Hu et al. (2023)	0.624	0.541	0.477	0.422	0.337	0.645	1.893
Yung et al. (2024)	0.705	0.633	0.572	0.517	0.381	0.651	1.804
Cho et al. (2024)	0.704	0.633	0.575	0.520	0.381	0.653	1.832
Marhuenda et al. (2025)	0.716	0.647	0.590	0.537	0.389	0.670	2.119
Lu et al. (2024)	0.710	0.636	0.580	0.530	0.395	0.736	2.409
ALIGN-VQA	**0.653**	**0.578**	**0.518**	**0.466**	**0.480**	**0.739**	**2.427**

Table 5.2

Ablation studies isolating the impact of the Top-k selection threshold, CIDA, and GATE.

Top-k	CIDA	GATE	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	BERT-F1
12	×	×	0.566	0.493	0.437	0.386	0.414	0.635	1.757	0.833
12	×	✓	0.625	0.551	0.493	0.442	0.416	0.624	1.853	0.844
12	✓	×	0.623	0.553	0.499	0.449	0.459	0.676	2.118	0.849
12	✓	✓	0.636	0.541	0.502	0.450	0.481	0.700	2.258	0.952
49	×	×	0.600	0.527	0.470	0.420	0.416	0.619	1.923	0.942
49	✓	×	0.669	0.591	0.532	0.480	0.462	0.665	2.089	0.947
49	×	✓	0.566	0.500	0.449	0.406	0.411	0.611	1.915	0.941
49	✓	✓	0.631	0.555	0.496	0.445	0.480	0.713	2.226	0.948
16	×	×	0.215	0.165	0.134	0.108	0.306	0.490	1.383	0.816
16	✓	×	0.643	0.565	0.504	0.453	0.457	0.660	2.172	0.875
16	×	✓	0.661	0.577	0.511	0.453	0.454	0.661	1.956	0.887
16	✓	✓	**0.653**	**0.578**	**0.518**	**0.466**	**0.480**	**0.739**	**2.427**	**0.953**

Table 5.3

Evaluation scores across different clinical question types using the best performing ALIGN-VQA model (Top-16, CIDA-active, Gate-active).

Question	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr
difference	0.663	0.589	0.528	0.473	0.659	0.624	2.177
presence	0.849	0.009	0.001	0.000	0.424	0.849	2.123
abnormality	0.605	0.520	0.518	0.414	0.367	0.599	1.904
location	0.616	0.489	0.359	0.275	0.540	0.668	2.496
level	0.653	0.007	0.001	0.000	0.395	0.598	1.984
view	0.979	0.965	0.098	0.001	0.649	0.982	3.343
type	0.706	0.008	0.001	0.000	0.360	0.720	1.900
overall	**0.653**	**0.578**	**0.518**	**0.466**	**0.480**	**0.739**	**2.427**

the ALIGN-VQA model equipped with CIDA. The results drawn from the comprehensive ablation studies demonstrate a severe vulnerability in direct subtraction methods.

When observing the model operating with a token pool of 16, the baseline model lacking both CIDA and GATE completely collapsed, achieving CIDEr, Oliveira dos Santos et al. (2021), of just 1.383 and BLEU-4, Papineni et al. (2002), of 0.108. However, when the CIDA module was activated to semantically align the features prior to comparison, performance surged dramatically. The score of CIDEr improved by +0.789 to reach 2.172, while BLEU-4 improved to 0.453. This massive performance delta mathematically validates Hypothesis 1, proving that raw spatial misalignment is a primary cause of catastrophic failure in longitudinal VQA, and that CIDA successfully reconstructs the features to enable accurate temporal comparison.

5.2 Results Corresponding to GATE

The second research question asks how the application of a learnable residual gating mechanism impacts the model's ability to capture precise pathological deviations. The author hypothesized that filtering out non-pathological noise, such as changes in patient posture or scanner contrast, from the raw residual features would yield higher diagnostic accuracy.

The ablation results on the specific difference question subset strongly support this hypothesis. Even with active CIDA module, processing raw, un-gated residuals resulted in a CIDEr score of 2.081. When GATE was introduced to selectively suppress static anatomy and acquisition artifacts, the CIDEr score increased to 2.177, and the BERT-F1, Zhang* et al. (2020), score rose from 0.917 to 0.929. This impact is even more pronounced on the overall dataset evaluation. Activating the gating mechanism alongside CIDA pushed the model to its absolute peak performance: a CIDEr score of 2.427 and a ROUGE-L, C.-Y. Lin (2004), of 0.739. By comparing the gated vs. un-gated architectures, the author confirms that forcing the language decoder to process noisy subtraction maps degrades clinical accuracy, whereas the gating mechanism successfully isolates the true pathologies.

5.3 Results Corresponding to QDT

The last research question evaluates whether selectively pooling temporal changes via the QDT impacts the model's targeted reasoning and overall efficiency. The author hypothesized that because the vast majority of pixels in a longitudinal chest X-ray remain unchanged, a sparse set of highly relevant tokens is sufficient—and often superior—for answering clinical questions. The author tested the model across three different token densities.

Processing the denser 49 tokens yielded a CIDEr score of 2.226. Pruning the features down to 16 tokens improved the CIDEr score to 2.427. Over-pruning to 12 tokens caused performance to drop to 2.258. These results are highly significant. They prove that retaining too much visual information by processing dense features actually harms the model by introducing distracting, irrelevant visual features. The QDT module at Top-16 perfectly balances computational efficiency with diagnostic accuracy, effectively pruning away the background and forcing the model to focus only on the region of interest dictated by the question.

5.4 Further Analysis

5.4.1 Comparison with State of the Art Models

To validate the overall architecture, ALIGN-VQA was benchmarked against leading state-of-the-art Medical VQA models, including EKAID, RegioMix, PLURAL, VED, and ReAI. ALIGN-VQA achieved the highest overall performance across the most stringent clinical metrics. Most notably, ALIGN-VQA achieved a CIDEr score of 2.427, outperforming the closest baseline (ReAI at 2.409) and vastly exceeding traditional architectures like EKAID (1.893). Furthermore, ALIGN-VQA secured the highest ROUGE-L score (0.739), indicating that the architecture is superior at preserving the correct chronological and relational structure of the diagnostic narrative compared to existing models.

5.4.2 Analyzing Distribution Across Question Types

Because clinical queries vary drastically in complexity, we further analyzed the performance of the best model (Top-16, CIDA-active, Gate-active) across distinct question categories. The model showed exceptional performance on simpler, highly structured tasks such as view questions (CIDEr: 3.343, BLEU-1: 0.979) and location questions (CIDEr: 2.496).

5.4.3 Human Evaluation and Error Analysis

All test samples are manually evaluated beyond Exact Match (EM). While EM was 51.1%, manual assessment showed 50% fully correct, 30% partially correct, and 20% incorrect predictions. Partial errors mainly involved incomplete specification or minor lexical variations. Performance was strongest for structured categories (type, view: 93.7% correct), whereas more complex categories (abnormality, difference) showed greater variability (55.8% fully correct, 25.0% partially incorrect, 18.8% incorrect), reflecting the challenge of fine-grained pathology characterization and comparative reasoning. Importantly, no systematic localization failures were observed, and most errors resulted from partial specification rather than clinically misleading predictions.

CHAPTER 6
DISCUSSION

The development and evaluation of the ALIGN-VQA architecture expose fundamental truths about how artificial intelligence must approach longitudinal medical imaging. For years, the prevailing methodology in Difference Visual Question Answering (Diff-VQA) has relied on the implicit assumption that deep neural networks can naturally discern temporal changes through raw feature subtraction or concatenation. The findings of this thesis categorically dismantle that assumption. This chapter synthesizes the empirical results, examines the unexpected behaviors of the model, justifies the critical architectural decisions that drove the success, and candidly addresses the boundaries of the current approach.

6.1 Revisiting Research Questions

The central thesis of this work was that longitudinal VQA requires explicit spatial alignment, targeted noise filtration, and question-guided reasoning. Across the three research questions, the empirical data not only validated these claims but revealed unexpected dynamics regarding how vision-language models process temporal medical data.

6.1.1 Cross Image Difference Attention

53

The author fully expected that semantically aligning the reference image to the current image would improve performance. However, the magnitude of the baseline's failure without difference attention module was highly unexpected. When processing unaligned scans using direct subtraction, the model's linguistic output almost entirely collapsed, dropping to a BERT-F1, Zhang* et al. (2020), of 0.818 and CIDEr, Oliveira dos Santos et al. (2021), of 1.383. This catastrophic drop revealed a critical insight. Without explicit spatial alignment, generative language models do not simply make minor diagnostic errors. They lose the ability to ground their language in visual reality, resorting to fragmented, incoherent guesses. The module proved that spatial correspondence is not merely a performance booster although it is a fundamental prerequisite for coherent longitudinal reasoning.

6.1.2 Learnable Residual Gating

The author hypothesized that explicitly filtering the raw subtraction features would remove static background noise and improve diagnostic accuracy. However, what the author did not expect was the emergent semantic sophistication of the gate itself. The author initially assumed the gating mechanism would function primarily as a rigid spatial mask, simply zeroing out unchanged bones or static medical hardware. Qualitative analysis of the gated features revealed that the model learned to handle highly complex, dynamic non-pathological variations. For instance, it actively suppressed visual artifacts caused by differing levels of patient inspiration, deep of a breath the patient took between the two scans. As a profound realization, the network didn't just learn to mask unchanged pixels although it learned a deep clinical heuristic for successfully distinguishing between a shift in lung volume and a genuine change in lung fluid/opacity.

6.1.3 Question Guided Tokens and Targeted Reasoning

The author anticipated that pruning tokens would make the model more computationally efficient although the author expected a slight trade-off in diagnostic accuracy due to lost information. The results completely subverted this expectation. The sparse Top-16 token configuration actually outperformed the dense Top-49 configuration CIDEr 2.427 vs. 2.226. This is a profound finding. It suggests a paradox, less is more, in medical VQA: dense visual features act as a distraction. Because the vast majority of a chest X-ray remains unchanged between visits, feeding the language decoder 49 tokens overwhelms its cross-attention layers with static, irrelevant anatomy. By forcing the Question Guided Tokenizer (QDT) to aggressively prune 70 percent of the image and pass forward only 16 highly relevant tokens, it inadvertently improved the decoder's focus. However, the author also found the lower bound, which pushes the sparsity to Top-12 tokens resulted in a performance drop, indicating the threshold where critical contextual margins around a pathology are destroyed.

6.2 Comparing with Past Work

ALIGN-VQA successfully achieved the primary claim, which is a lightweight, highly accurate architecture capable of interpreting disease progression while filtering out non-pathological noise. ALIGN-VQA currently outperforms state-of-the-art models like Lu et al. (2024) and Cho et al. (2024), achieving peak structural and semantic scores. However, ALIGN-VQA did not yet achieve flawless reasoning on sub-pixel or highly abstract relational questions involving complex spatial reasoning across multiple intersecting pathologies. The ultimate goal is to develop a fully autonomous, zero-shot longitudinal reasoning engine that matches the contextual awareness of an experienced radiologist, a system capable of tracking disease trajectories over not just two, but dozens of sequential visits. We are currently one major step closer to this goal; by solving the spatial misalignment and noise bottlenecks, the author has laid the necessary groundwork. The remaining distance requires shifting from static two-image comparison to dynamic, multi-hop temporal graph reasoning.

6.3 Experimental Decisions

The success of ALIGN-VQA is heavily predicated on specific, deliberate experimental and architectural design choices. Rather than relying on standard transformer pipelines, the author engineered the workflow to mirror the cognitive process of a radiologist.

6.3.1 Architectural Sequencing

A major architectural decision was the strict sequential ordering of the modules. In early ideation, one might assume that pruning tokens first and then aligning them would save compute. The author deliberately chose to align the dense features first, gate the dense residuals, and only then apply the QDT to prune the tokens.

This decision was critical to the results. Spatial alignment requires full global context as the network must see the clavicles, the diaphragm, and the rib cage to calculate the semantic transformation matrices. If the model had pruned the tokens prior to alignment based on the text question, the CIDA module would be starved of the global anatomical landmarks required to align the images. Similarly, the Learnable Residual Gate needs the surrounding healthy tissue to establish a baseline for normal noise before it can filter it. By placing the QDT at the very end of the visual encoder pipeline, the author ensured that the tokens selected for the language decoder were already perfectly aligned and scrubbed of acquisition artifacts. This design choice directly enabled the high CIDEr scores observed in the ablation studies.

6.3.2 The Selection of NLG Metrics for Clinical Evaluation

A second crucial experimental decision was the heavy reliance on CIDEr and ROUGE-L, C.-Y. Lin (2004), alongside traditional BLEU, Papineni et al. (2002), metrics. In standard computer vision, accuracy or simple BLEU scores are often deemed sufficient.

The author explicitly designed the evaluation around CIDEr because medical reporting is a long-tail linguistic distribution. In a clinical setting, a generic phrase like "The lungs are clear" appears thousands of times in the dataset, whereas a critical phrase like "mild pneumothorax has worsened" appears rarely. BLEU treats all words equally. If a model defaults to safe, high-frequency phrasing, it can artificially inflate its BLEU score while missing the actual disease. The author chose CIDEr because its Term Frequency-Inverse Document Frequency (TF-IDF) weighting heavily penalizes models that guess generic templates and explicitly rewards the model for generating rare, highly specific diagnostic keywords. By utilizing CIDEr as the primary metric for diagnostic accuracy, the author forced the gating mechanism to genuinely learn the difference between background noise and true, rare pathological evolution, rather than allowing the model to cheat the evaluation through language priors.

6.4 Limitation and Future Work

While ALIGN-VQA demonstrates state-of-the-art performance in longitudinal reasoning, it is constrained by a fundamental structural limitation regarding how it perceives multi-dimensional space.

6.4.1 Limitation

The most concrete limitation affecting the validity of this architecture is its absolute reliance on 2-Dimensional feature projections. While the CIDA module is highly effective at correcting 2D spatial misalignments, such as planar translations, vertical slouching, or varying inspiration volumes, it cannot semantically reconstruct 3D out-of-plane rotations.

In clinical reality, a patient might be slightly rotated laterally relative to the X-ray detector in their current visit compared to their reference visit. Because a chest X-ray is a 2D projection of a 3D volume, an out-of-plane rotation fundamentally alters the overlapping densities of the ribs, heart, and lung parenchyma. As the CIDA module operates on 2D semantic maps, it will attempt to forcefully align these features, which may warp or distort the underlying pathological representation. Consequently, the Learnable Residual Gate might misinterpret the sudden appearance of a rotated rib shadow as a "new opacity," or conversely, gate out a true lesion because the rotational alignment obscured it. This 3D-to-2D projection ambiguity represents a hard ceiling on the accuracy of any purely 2D longitudinal subtraction model, including ALIGN-VQA.

6.4.2 Future Work

Future iterations of ALIGN-VQA should abandon 2D constraints and simultaneously process paired posteroanterior and lateral views. By constructing a multi-view temporal graph, the architecture could triangulate the Z-axis depth of a pathology. If the CIDA module is extended to cross-reference alignment matrices between posteroanterior and lateral views, the model could mathematically differentiate between an out-of-plan patient rotation, which affects both views systematically, and a genuine volumetric tumor growth, effectively nullifying the current core limitation.

Currently, the Learnable Residual Gate filters visual noise before the model processes the clinical question. While this works well for general noise reduction, what constitutes "noise" is often dependent on the clinical context. For example, surgical hardware is "noise" if the question asks about pneumonia, but it is the "signal" if the question asks about a pacemaker lead. Future work should implement a Question-Conditioned Gate, where the text embedding of the clinical query is injected directly into the gating bottleneck. This would allow the gate to dynamically redefine its filtration parameters on a per-query basis, resulting in even higher efficiency and diagnostic precision.

CHAPTER 7
CONCLUSION

The primary objective of this research was to address the inherent vulnerabilities in current Medical Diff-VQA models—specifically their susceptibility to spatial misalignments, their inability to filter out non-pathological acquisition noise, and their lack of targeted reasoning. Through the development of the ALIGN-VQA architecture, we have demonstrated that accurate longitudinal reasoning is not achieved through "more data" or "larger models," but through structural alignment and precise feature selection.

7.1 Main Findings

The empirical evaluation on the Medical-Diff-VQA benchmark yielded several critical insights. Most notably, ALIGN-VQA achieved a state-of-the-art performance, outperforming existing models and demonstrating a superior ability to preserve the chronological clinical narrative. The ablation studies provided mathematical proof of the hypotheses. The model reached its peak performance when all three modules worked in concert, highlighting the necessity of the Align $\rightarrow$ Filter $\rightarrow$ Focus pipeline. Furthermore, the qualitative heatmaps confirmed that the model's reasoning is explicitly grounded in the visual evidence of the scans, moving the model away from the "black box" nature of conventional generative AI.

7.2 Final Reflection

The path toward truly trustworthy medical AI requires models that do not just detect changes but understand the context of change over time. While the current iteration of ALIGN-VQA is limited to 2D projections and pairs of images, it provides a rigorous blueprint for the next generation of longitudinal diagnostic tools. Ultimately, this work serves as a reminder that in the high-stakes domain of radiology, precision is a product of focus. By filtering out the noise of the healthy body, the model can see the features of the disease to speak more clearly. As moving toward a future where AI serves as a reliable co-pilot for clinicians, architectures like ALIGN-VQA will be fundamental in ensuring that these systems are not just fast, but clinically grounded, structurally accurate, and fundamentally robust.

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APPENDICES

VITA

This section presents a short description of the educational and professional achievements of the student.